

- Outer $x=10$ is hidden (shadowed).
- After inner block ends:-
 - Inner table is deleted.
 - Outer x becomes visible again.

5) Run-time storage allocation refers to how memory is organized and assigned to variables and functions during program execution. There are three main types:-

a) Static Allocation:-

- Memory is allocated at compile time.
- Size and location are fixed.

Characteristics:-

- Cannot be changed at runtime.
- Exists throughout program execution.

Used in:-

- Global variables.
- Static variables.

Advantages:-

- Fast access.
- No runtime overhead.

Disadvantages:-

- No flexibility.
- Memory wastage possible.

Eg:- `int x=10;`